RR

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ROAD SAFETY INTELLIGENCE

Road Collision Risk & Spatial Inequality

Where is road risk high after exposure, and where is collision burden socially concentrated?

↻ Reset filters

YEAR: 2025

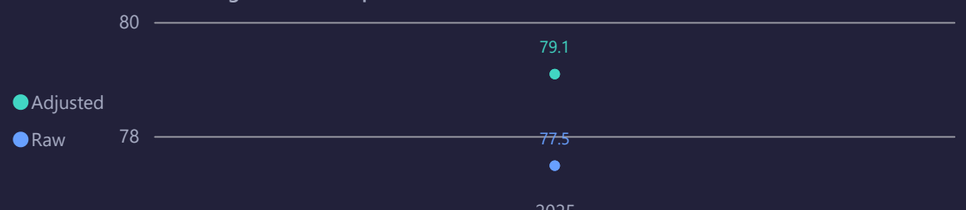
RISK METRIC: Adjusted expected KSI

ⓘ Resident-normalised rates show community burden, not traffic exposure.

Reported collisions	Adjusted KSI / bn vehicle miles	Deprivation burden ratio	Traffic-exposure coverage
101,525	79.1	1.98x	99.97%
	▲ 1.5% vs previous year		

NATIONAL TREND

KSI risk after accounting for traffic exposure



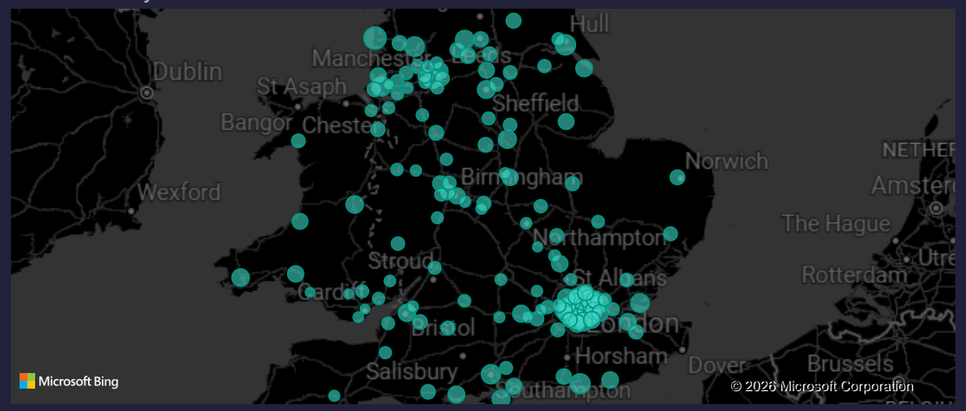
DECISION SIGNAL

Two problems require two intervention tracks

2.03×	Five-year collision burden: most deprived versus least deprived neighbourhoods.
31.7	Adjusted expected KSI per 100 rural collisions in the selected period.

GEOGRAPHIC DISTRIBUTION

Local-authority KSI risk · Bubble size = selected metric



INVESTIGATION QUEUE

Highest local-authority risk · ≥100 collisions in selected period



● Reviewed data
2021 -2025

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Separate community burden from traffic exposure before setting place-based priorities.

↻ Reset filters

YEAR

2025

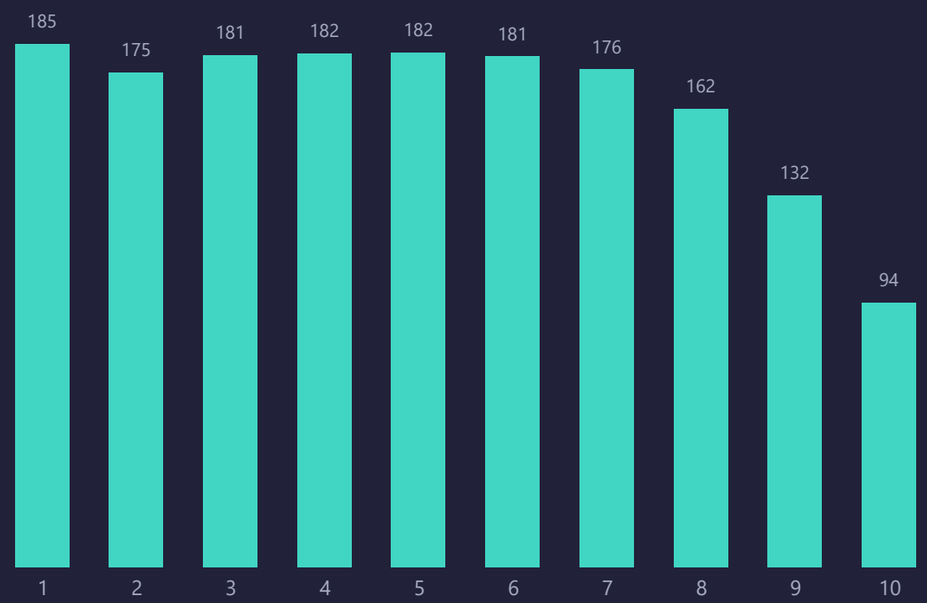
SPATIAL INEQUALITY

The deprivation gap reflects collision occurrence rather than conditional severity.

ⓘ Resident-normalised rates show community burden, not traffic exposure.

Collision burden by deprivation decile

Collisions per 100k residents per year



Annual most-to-least deprived gap

Zoomed scale · 1.00 = parity



INTERPRETATION

The inequality signal is persistent, but it is not an exposure-adjusted road-risk estimate.

Use this view to prioritise neighbourhood diagnosis and place-based prevention. Do not use it to claim that deprivation causes collisions or that residents themselves generated the traffic exposure.

● Reviewed data
2021 -2025

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ROAD SAFETY INTELLIGENCE

Road Collision Risk & Spatial Inequality

Compare collision consequences across road environments without implying causality.

↻ Reset filters

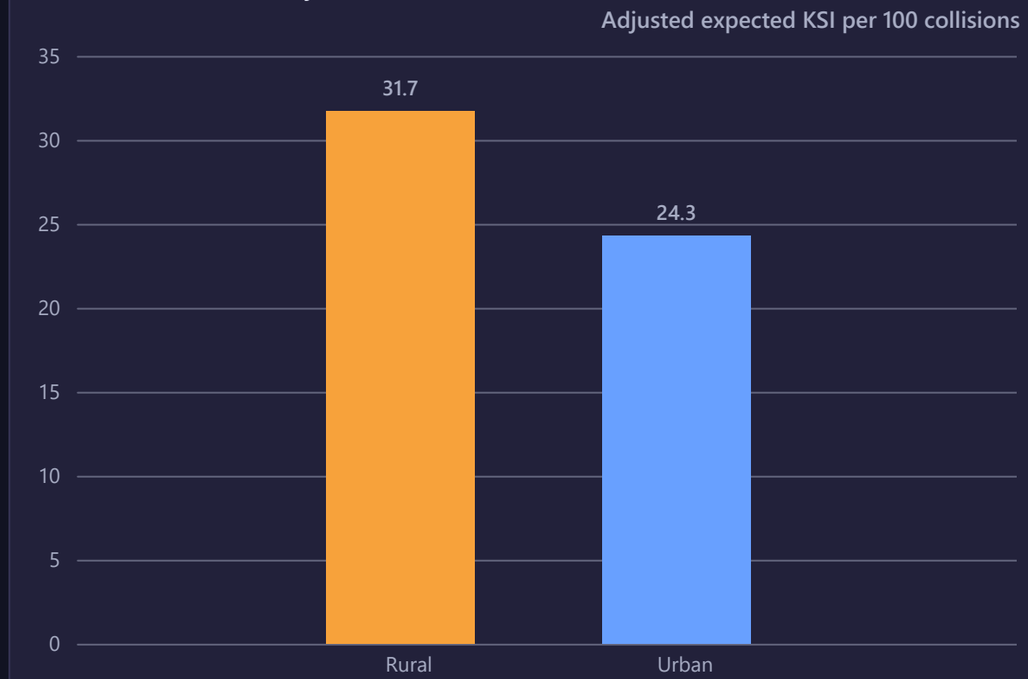
YEAR

2025

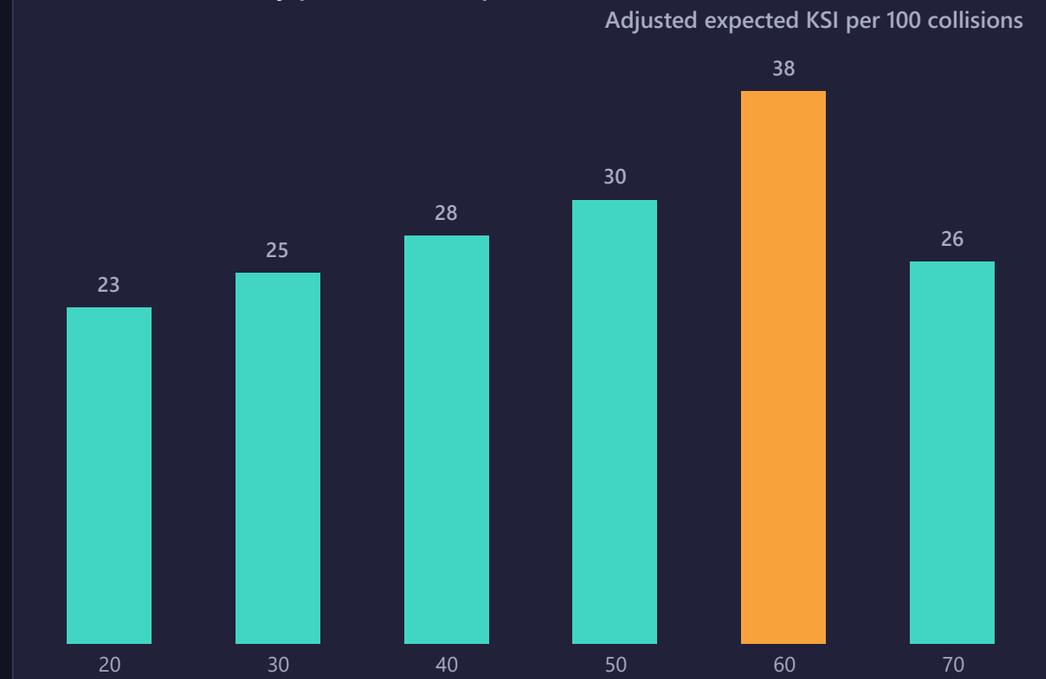
ROAD RISK & SEVERITY

Context changes the consequences of a collision

Urban vs rural severity



Conditional severity peaks on 60 mph roads



INTERPRETATION

Speed limit is informative, but not a sufficient causal explanation.

Conditional severity peaks on 60 mph roads and falls on 70 mph roads. Road class, traffic separation, junction context and road-user mix should be modelled jointly before recommending a specific intervention.

• Reviewed data
2021 -2025

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ROAD SAFETY INTELLIGENCE

Road Collision Risk & Spatial Inequality

Exposure coverage, matching limits and measurement sensitivity beside the analysis.

↻ Reset filters

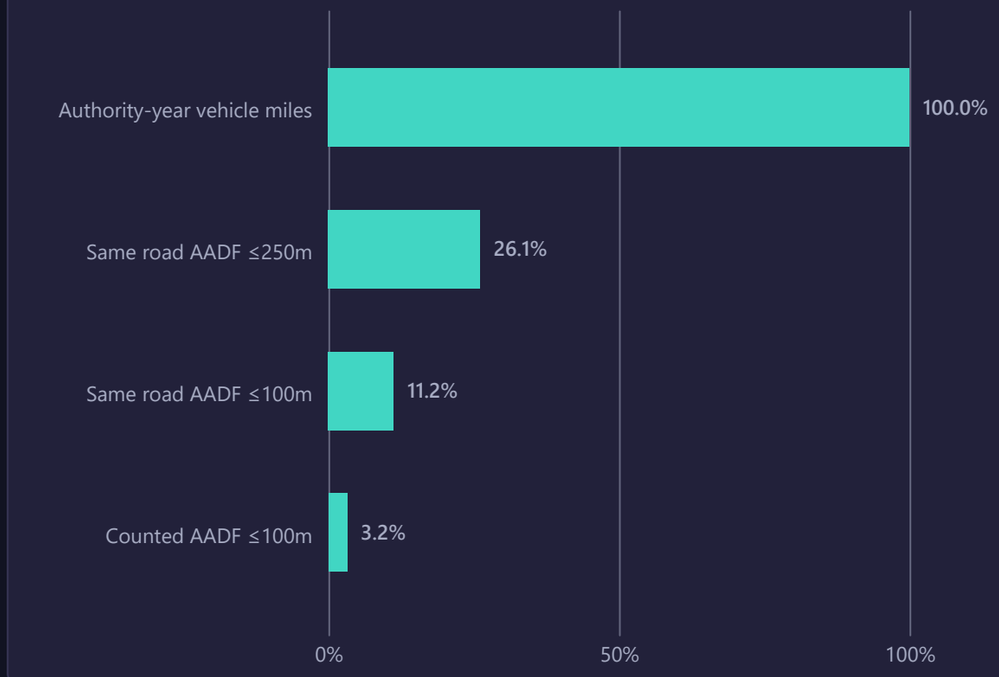
YEAR

2025

DATA QUALITY & SENSITIVITY

Make analytical confidence visible beside the result

Exposure scenario coverage



MODEL GUARDRAILS

Local-authority traffic coverage

99.97% — Authority-year vehicle miles

England-to-IoD join

100% — Applicable English collision records

AADF ≤100m coverage

11.2% — Eligible numbered-road collisions

Raw vs adjusted rank correlation

0.991 — 2025 Spearman ρ

LIMITATION

AADF point matches are a sensitivity diagnostic, not a link-level absolute risk denominator.

Do not use the AADF coverage scenarios to publish absolute road-link risk rates until road-segment geometry matching is available.

● Reviewed data
2021 -2025